

Transient Activation Steering Research Notebook

This notebook reconstructs the FeatureScalpel MVP for inference-time mechanistic interventions using Sparse Autoencoders (SAEs).

Unlike fine-tuning, no model parameters are trained or modified. Both GPT-2 Small and the SAE are downloaded as pretrained artifacts and cached to Google Drive to avoid repeated downloads across Colab sessions.

At the end of this section we should have:

- Google Drive mounted
- GPT-2 Small available
- Pretrained SAE available
- Both cached for future sessions

```
1 from google.colab import drive
2
3 drive.mount("/content/drive")
```

Mounted at /content/drive

```
1 !pip -q install transformer-lens sae-lens
```

```
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
145.1/145.1 kB 6.5 MB/s eta 0:00:00

Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
1.1/1.1 MB 31.9 MB/s eta 0:00:00
312.2/312.2 kB 25.9 MB/s eta 0:00:00
56.6/56.6 kB 6.1 MB/s eta 0:00:00
26.8/26.8 MB 84.9 MB/s eta 0:00:00
215.2/215.2 kB 22.1 MB/s eta 0:00:00
233.3/233.3 kB 20.2 MB/s eta 0:00:00
503.9/503.9 kB 40.7 MB/s eta 0:00:00
10.3/10.3 MB 144.0 MB/s eta 0:00:00
42.6/42.6 kB 4.0 MB/s eta 0:00:00
62.8/62.8 kB 6.3 MB/s eta 0:00:00
52.4/52.4 kB 4.2 MB/s eta 0:00:00
298.4/298.4 kB 27.6 MB/s eta 0:00:00
236.9/236.9 kB 23.7 MB/s eta 0:00:00

Building wheel for transformers-stream-generator (pyproject.toml) ... done
Building wheel for py2store (pyproject.toml) ... done
```

Next we define a persistent cache directory inside Google Drive.

Any downloaded weights will be stored here.

Future notebook sessions will reuse these files instead of downloading them again.

```
1 from pathlib import Path
2
3 CACHE_DIR = Path("/content/drive/MyDrive/Transient_Activation_Steering")
4 CACHE_DIR.mkdir(parents=True, exist_ok=True)
5
6 MODEL_DIR = CACHE_DIR / "gpt2_small"
7 SAE_DIR = CACHE_DIR / "sae_layer8"
8
9 print(CACHE_DIR)
```

/content/drive/MyDrive/Transient_Activation_Steering

We now load GPT-2 Small and the pretrained SAE.

On the first execution they are downloaded.

On later executions they are loaded directly from Google Drive.

This caching step does **not** retrain the model.

```
1 import torch
2 from transformer_lens import HookedTransformer
3 from sae_lens import SAE
```

```
1 device = "cuda" if torch.cuda.is_available() else "cpu"
2
3 print(device)
```

cpu

```
1 model = HookedTransformer.from_pretrained(
2     "gpt2",
3     device=device
4 )
5
6 sae, cfg_dict, sparsity = SAE.from_pretrained(
7     release="gpt2-small-res-jb",
8     sae_id="blocks.8.hook_resid_pre",
9     device=device
10 )
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:138: UserWarning:
Error while fetching `HF\_TOKEN` secret value from your vault: 'Requesting secret HF_TOKEN timed out. Secrets can only be fetched
You are not authenticated with the Hugging Face Hub in this notebook.
If the error persists, please let us know by opening an issue on GitHub (https://github.com/huggingface/huggingface_hub/issues/)
warnings.warn(
config.json: 100% 665/665 [00:00<00:00, 107kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
model.safetensors: 100% 548M/548M [00:01<00:00, 1.37GB/s]
Loading weights: 100% 148/148 [00:00<00:00, 6095.59it/s]
generation_config.json: 100% 124/124 [00:00<00:00, 23.7kB/s]
tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 4.56kB/s]
vocab.json: 100% 1.04M/1.04M [00:00<00:00, 7.28MB/s]
merges.txt: 100% 456k/456k [00:00<00:00, 3.34MB/s]
tokenizer.json: 100% 1.36M/1.36M [00:00<00:00, 23.8MB/s]
Loaded pretrained model gpt2 into HookedTransformer
cfg.json: 100% 1.27k/1.27k [00:00<00:00, 203kB/s]
blocks.8.hook_resid_pre/sae_weights.safe(...): 100% 151M/151M [00:01<00:00, 769MB/s]
blocks.8.hook_resid_pre/sparsity.safeten(...): 100% 98.4k/98.4k [00:00<00:00, 154kB/s]
/usr/local/lib/python3.12/dist-packages/sae_lens/saes/sae.py:254: UserWarning:
This SAE has non-empty model_from_pretrained_kwargs.
For optimal performance, load the model like so:
model = HookedSAETransformer.from_pretrained_no_processing(..., **cfg.model_from_pretrained_kwargs)
warnings.warn(
/tmp/ipykernel_423/3360701761.py:6: DeprecationWarning: Unpacking SAE objects is deprecated. SAE.from_pretrained() now returns d
sae, cfg_dict, sparsity = SAE.from_pretrained(

